Rahul Chandra Padamuttam

Ongole, Andhra Pradesh, India | rahulpc1981@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION & TECHNICAL STACK

National Institute of Advanced Manufacturing Technology (NIAMT), Ranchi *Jan 2024 – Apr 2028*
B.Tech in Electronics and Computer Engineering **CGPA: 7.25/10.0**

- **Languages & Core Systems:** C++, Python, Java, C, Data Structures & Algorithms, Object-Oriented Programming (OOP), AST Parsing, Linux, Multithreading
- **SRE, Security & Cloud:** Site Reliability Engineering (SRE), FastAPI, OpenTelemetry, SigNoz, ClickHouse, Docker, GitHub Actions CI/CD, Azure, Oracle Cloud (OCI)
- **Robotics, AI & Frameworks:** MuJoCo, PyQt6, OpenCV, TensorFlow/Keras, Scikit-Learn, Pandas, NumPy, DeepFace, MAVLink/ArduPilot SITL, Power BI (DAX/M)

EXPERIENCE

Maulana Azad National Institute of Technology (MANIT), Bhopal | *Research Intern (Robotics Software)* | **[GitHub]** *Jun 2026*

- Engineered an **11,847-LOC** locomotion control and kinematic analysis suite (28 modules) for a **35.0 kg, 12-DOF** floating-base humanoid biped in MuJoCo running symplectic physics integration and closed-loop PD torque control at **1,000 Hz** ($\Delta t = 1.0\,\text{ms}$) with **100 Hz** 14-channel telemetry harvesting.
- Benchmarked discrete Finite State Machine (FSM) keyframes against continuous Central Pattern Generator (CPG) phase manifolds, proving twice-differentiable C^2 limit cycles eliminate C^0 touchdown kinetic shock and extend episode walking survival **20.4**× (**6.13 s** → **125.0+ s** across 12,501 samples) within a ± 0.05 rad pitch envelope.
- Derived a **Smoothed Deadband IMU Reflex** to suppress actuator micro-chattering (reducing spectral jerk density from $10^7 \rightarrow 10^4$ PSD, eliminating **+215%** structural fatigue) and quantified late-stance parasitic braking work (**1.73 J** → **2.68 J**; peak braking power -82.4 W vs. -130.0 W).

Microsoft Elevate AICTE (Remote) | *Power BI & Data Observability Intern* | **[GitHub]** *Feb 2026 – Mar 2026*

- Architected an enterprise star-schema Quick Commerce observability model compressed to **28.8 MB** in Power BI, automating ingestion via Power Query (M) and engineering **5 core DAX KPI pipelines** across **14 visual containers** to benchmark INR 80.8M+ Swiggy Instamart order volume and JioMart delivery velocity.

Code/Forge, NIAMT Ranchi (On-Site) | *Core Member – Community of Robotics and Automation* *Jan 2026 – Present*

- Spearheaded systems architecture reviews, C++/Python robotics workshops, and multi-drone SITL simulation mentoring for 50+ undergraduate engineers.

SYSTEMS, DEVSECOPS & APPLIED AI PROJECTS

ArcP – Polymorphic Radial Desktop Firewall & Zero-Disk DevSecOps Pipeline | **[GitHub]** *Python · PyQt6 · WinRT · AST*

- Architected an **8,542-LOC** frameless 60-FPS radial security firewall across 30 modular components, achieving a **377.58 ms** cold start, a **53.08 MB RSS** zero-disk memory footprint, and **31/31 passing unit tests** (100% compliance) in headless CI/CD.
- Engineered the thread-safe **TokenVault** bidirectional redaction engine across **9 secret/PII signatures** with 600s TTL eviction, sustaining **103,538 ops/sec** at **0.00966 ms** (9.66 μs) for Shannon entropy detection ($H(X) > 4.2$, beating the 1.0 ms SLA by **103**×) and **6,838 ops/sec** (**0.1462 ms**) for full payload scrubbing.
- Built the **MimeRouter** magic-byte classifier routing 7 operational modes at **95,259 ops/sec** (**0.01050 ms** latency, **4.3**×–**5.4**× **faster** than disk stat checks), paired with **BinaryOps** AST security auditing at **5,251 ops/sec** (**0.1904 ms**) and native WinRT in-memory neural OCR (**984.24 ms**).

OmniSRE – Autonomous Closed-Loop AI Site Reliability Engineer | **[GitHub]** *FastAPI · OTel · SigNoz*

- Built a **6-container** self-healing SRE platform (**815 LOC** across Investigator, Notifier, and Healer services) instrumenting microservices with OpenTelemetry, ingesting distributed traces into ClickHouse via SigNoz, and executing sub-500 ms sliding-window span queries upon anomaly detection.
- Coupled **Gemini 1.5 Flash** root-cause synthesis with an asynchronous Telegram Human-in-the-Loop kill switch and direct Docker daemon container actuation, slashing P99 latency by **98.9%** (**3,420 ms** → **38 ms**), suppressing HTTP 500 chaos failures from **74.2%** → **0.0%**, and cutting MTTR from **45 min** → **38 s**.

UIDAI Aadhaar Lifecycle ETL & District Anomaly Detection Pipeline | **[GitHub]** *Python · Pandas*

- Engineered a 3-stage inner-join ETL pipeline processing **117M+ Aadhaar lifecycle records** (69M biometric and 48.8M demographic updates across 36 States/UTs) on composite geographical keys, isolating **300,000+ district-level anomalies** (Adult/Youth > 2.0) and mandatory age 5/15 biometric drop-offs.

Real-Time Facial Emotion CNN & Embedded IoT Gas Mitigation System | **[GitHub: CNN]** | **[GitHub: IoT]** *C++ · Keras · OpenCV*

- Deployed a **2.74 MB** Keras CNN trained on **35,887 FER-2013 images** (48×48, 7 classes) achieving **<35 ms** frame inference (∼**30 FPS**) in an OpenCV Haar Cascade GUI; programmed an embedded C++ Arduino Uno R3 gas mitigation rig interfacing MQ-5 sensors, SIM800A GSM alerts, and automated servo regulator shutoff.

CERTIFICATIONS

- **Cloud & Architecture:** Oracle Cloud Infrastructure (OCI) Certified Architect Associate (Aug 2026 – Aug 2028) | OCI Foundations Associate | Microsoft Certified: Azure Fundamentals, Copilot & Cloud Admin
- **AI, Data & Security:** Microsoft Elevate AI & Machine Learning | Microsoft SOAR AI | Cisco/IBM AI Fundamentals | IBM Python 101 | HP LIFE Data Science